Fiscal Year:	FY 2015	**Task Last Updated:**	FY 10/25/2014
PI Name:	Turker, Mitchell Ph.D.		
Project Title:	The Relation Between Mutagenesis and Genomic Instability After Particle Exposure In Vivo		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	97239-3011	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 Space Radiobiology NNJ13ZSA001N
Start Date:	01/16/2014	**End Date:**	01/15/2017
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Grygoryev, Dmytro (Oregon Health & Science University) Thayer, Mathew (Oregon Health & Science University) Kronenberg, Amy (Lawrence Berkeley National Laboratory)		
Grant/Contract No.:	NNX14AC97G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Astronauts in the space environment are at risk for a variety of health-related concerns from exposure to accelerated ions (fast moving atoms that have lost their electrons) that can pass through the spacecraft hull and interact with astronauts' cells within the body. These particles are released from the sun in solar particle events (SPE) and exist as part of the Galactic Cosmic Ray (GCR) background radiation, although the types of ions and their energies can differ widely in SPE and GCR. A significant concern for astronaut health is an increased risk of cancer, which is linked to a variety of radiation exposures. The underlying cause of cancer, regardless of its source, is mutations in critical genes and ongoing gains, losses, and rearrangement of chromosomes. Collectively, these ongoing changes in the cell are termed genomic instability. The goal of the proposed work is to determine how frequently different types of accelerated ions found in the GCR cause mutations and genomic instability and to describe what these changes look like. To better extrapolate our results to potential harms to astronauts from space radiation, we will use a mouse model for the proposed work. This will ensure that the cell types of interest found in the spleen and kidney will be exposed to accelerated ions in their normal tissue environment. We will focus on accelerated ions found in space for which relatively little information is currently available. Successful completion of the proposed work will help in the design of long-term travel for humans in space and aid in the protection of space travelers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We are currently developing a new cytogenetic based method to identify radiation-induced mutations directly in cell nuclei. If we are successful, this method will apply to a number of research projects including those that do not involve radiation exposure.
Task Progress:	Task Book Report (year 1). The Relation Between Mutagenesis and Genomic Instability After Particle Exposure In Vivo (NNX14AC97G) Publications in last year: Grygoryev D, Dan C, Gauny S, Eckelmann B, Ohlrich AP, Connolly M, Lasarev M, Grossi G, Kronenberg A, and Turker MS. "Autosomal Mutants of Proton-Exposed Kidney Cells Display Frequent Loss of Heterozygosity on Non-Selected Chromosomes" Radiation Research. 2014 181:452–463. PMID: 24758577 (from predecessor grant, NNX10AC12G) Abstracts in last year: 1. Gwen Hyrciw, Dmytro Grygoryev, Anna Ohlrich, Christian Dan, Michael Lasarev, Stacey Gauny, Amy Kronenberg, and Mitchell Turker, Induction of Radiation Signature Mutations In Vivo with Charged Particles at Low Fluence Levels Presented at the 2014 NASA Human Research Program Investigators' Workshop (HRP 2014), Galveston, TX, Feb. 2014. 2. Dmytro Grygoryev, Anna Ohlrich, Michael Lasarev, Stacey Gauny, Amy Kronenberg, Mitchell S. Turker, Cell Death and Autosomal Mutation in Mouse Splenocytes In Vivo Caused by HZE ions, Presented at Radiation Research Annual Meeting, Las Vegas, NV, Sept. 2014. Work performed over the last year. 1. Oxygen ion exposed mice. We are using a mouse strain designed to measure mutations that form in the body to measure the frequency and types of mutations caused by accelerated charged particles found in space. These are atoms that are stripped of electrons, which is why they are positively charged. In October 2013, mice were exposed at Brookhaven National Laboratory (BNL) to oxygen (O) ions at doses of 0, 0.4, 0.8, and 1.6 Gy. A Gy (gray) is a unit that measures the absorption of radiation by the body. The numbers of mice exposed at each dose were 19 at 0 Gy, 28 at 0.4 Gy, 22 at 0.8 Gy, and 18 at 1.6 Gy. For each mouse, one kidney and the spleen were removed 3-5 months after exposure, cell suspensions were made, and the cells plated in tissue culture dishes to quantify mutant frequencies and determine cell viability. The cell viability data will be used to determine relative levels of lingering toxicity in the cells of each tissue as a result of exposure to the different doses of O ions. The mutant frequency data will be used to determine the shape of the dose response curve for the induction of mutants. 2. Our work during the last funding period with Fe ions (0.25, 0.5, 1, and 2 Gy), Ti ions (0.2 Gy, 0.4 Gy, and 1.4 Gy), and protons, (0.5, 1.0, and 4.0 Gy) suggests that we will observe a dose response curve for mutant frequency for the kidney cells, but not for the splenic T cells. This result, however, does not necessarily mean that mutants are not induced in the splenic T cells exposed to accelerated charged particles (see below). We have obtained mutant frequency and cloning efficiency data for the O ion exposed mouse kidney and spleen tissues over the last year and currently the results are awaiting statistical analyses. 3. During the last funding period we found that specific types of mutations in both the kidney and T cells can serve as signature mutations that can be used to detect radiation effects even at low doses. These mutations are large deletions within chromosomes and complex patterns of loss of regions of chromosomes called discontinuous loss (DL). For kidney cells we also measure chromosome translocations where parts of different chromosomes fuse, as an additional radiation signature mutation. Despite the lack of an increase in mutant frequency for the splenic T cells as the dose of radiation increases, we readily detected radiation signature mutations in these cells. Moreover, the signatures mutations were common at relatively low doses, but not detected at the highest doses tested. These results provide the basis for our expectation of observing radiation signature mutations in the O ion exposed mouse T cells even if we do not observe increased mutant frequencies. Radiation signature mutations were also observed in the mutant kidney cells from the same mice at all doses tested (not shown), suggesting that we will observe similar result with the kidney mutants from the O ion exposed mice. The work to identify the types of mutations will be performed in year 2. 4. During the last year, mutant cells were isolated from both kidney epithelial cells and splenic T cells for all O ion exposure conditions and DNA collected to identify the types of mutations that were caused by the exposure. The number of DNA preparations collected for mutant splenic T cells is as follows: 65 for 0 Gy, 107 for 0.4 Gy, 79 for 0.8 Gy, and 68 for 1.6 Gy. For mutant kidney epithelial cells we have collected 16 for 0 Gy, 46 for 0.4 Gy, 46 for 0.8 Gy, and 67 for 1.6 Gy. We also stored kidney mutant cells for future work to look at chromosome translocations. The mutation and chromosome work and will be performed during the next year. 5. Si ion exposed mice. In June 2014, 99 mice were exposed to Si (silicon) ions as follows: 0 (18 mice), 9 cGy (30 mice), 18cGy (27 mice), and 72 cGy (and 24 mice). We started sacrificing these mice in October 2014 and are currently setting up plates for the isolation of mutant splenic T cells and kidney epithelial cells. We are also currently breeding more mice for work at BNL in either spring or summer 2015. These mice will be exposed to Ca (calcium ions).

	6. Development of new method to detect radiation mutations in intact cells. One goal of current funding is to create a mutation assay to use directly on growing cells that would speed up the detection of radiation-induced mutations. To do this, we propose to use bacterial artificial chromosomes (BACs) to stain specific regions of the target chromosome to determine if any regions of the chromosome are missing in the mutant cells. We are currently validating the system by working with known mutants and staining metaphase spreads to ensure that each BAC recognizes the appropriate chromosome region and that translocation chromosomes could be identified. During the next year we will finish validating this method with known mutants and then test newly isolated mutants. This method has proven to be more expensive than anticipated, but we are still confident it will be a useful tool going forward, particularly for the splenic T cell mutants which cannot be expanded for as long as the kidney cell mutants.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
Abstracts for Journals and Proceedings	Hyrciw G, Grygoryev D, Ohlrich A, Dan C, Lasarev M, Gauny S, Kronenberg A, Turker M. "Induction of Radiation Signature Mutations In Vivo with Charged Particles at Low Fluence Levels." Presented at the 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. http://www.hou.usra.edu/meetings/hrp2014/pdf/3025.pdf , Feb-2014
Abstracts for Journals and Proceedings	Grygoryev D, Ohlrich A, Lasarev M, Gauny S, Kronenberg A, Turker MS. "Cell Death and Autosomal Mutation in Mouse Splenocytes In Vivo Caused by HZE Ions." Presented at the 60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014. 60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014. , Sep-2014